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A large-scale hybrid micro- and mesoscopic simulation approach for railway operation

Ihab Kaddoura^{a,*}, Merlin Unterfinger^a, Thomas Hettinger^a,
Christian Rakow^b, Marcel Rieser^b

^aSwiss Federal Railways (SBB), Corporate Development, Hilfikerstrasse 1, 3000 Bern, Switzerland

^bSimunto GmbH, Gubelstrasse 28, 8050 Zurich, Switzerland

Abstract

In this study, the existing agent-based simulation framework MATSim is extended towards a module for the simulation of trains and their interactions with the infrastructure. The proposed railway simulation module *railsim* offers several advantages over existing railway planning tools: *railsim* is open-source, easily adaptable, allows for a microscopic, mesoscopic or hybrid micro-mesoscopic simulation setup and is fully integrated in an existing simulation framework. Trains have a spatial dimension along several infrastructure segments. Trains accelerate and decelerate based on vehicle- and infrastructure-specific attributes as well as the available capacity ahead of the vehicle. The interaction of vehicles as well as the modeling of the infrastructure is adapted to the railway system. Trains are enabled to reserve infrastructure elements (fixed block segments, moving blocks), and reservation requests interact with each other. The illustrative Switzerland case study reveals that an overall unique capacity of 3 trains per section and 10 trains per station ensures most trains reach their destinations successfully, while lower capacities lead to insufficient infrastructure and train disruptions.

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* Corresponding author.

E-mail address: ihab.kaddoura@sbb.ch

1. Introduction and literature review

In most countries, the demand for rail transport is expected to increase significantly in the coming years. This is due to several effects: Transport policies which aim at a modal shift from motorized road traffic to rail, the increase in population, rising economic production, or changes in spatial structures. It is therefore increasingly important to account for the operational aspects of the rail system. One of the key elements is to identify and eliminate bottlenecks through infrastructural policies or operation concepts.

Today, transport planners are challenged by planning tools which require expert knowledge and human interaction. Most tools only provide assistance but do not allow for an automated feasibility check and evaluation of the railway schedule. The assistance is provided to the transport planner by visualizing the railway lines using so called time-distance diagrams. The planning process then requires to manually check if time-distance curves intersect and if yes, the infrastructure provides sufficient capacity, e.g., number of tracks, to pass each other. If not, the schedule is manually adjusted, and train departures need to be shifted to an earlier or later time which then requires again a check for potential conflicts. More sophisticated software tools such as OpenTrack (www.opentrack.ch) or Trenissimo (www.trenolab.com) provide additional functionality, including the detection of conflicts by simulating train operations on detailed track layouts. These tools offer a great potential for detailed quantitative studies, however, most of the existing software tools come with several drawbacks that impede their utility and applicability. These limitations include a closed-source nature, which makes the functionality inaccessible for detailed examination and modification by the user or research community. Moreover, their limited extensibility restricts the adaptation of the tools to future railway technologies or concepts, e.g., moving blocks. The strictly microscopic simulation approach and poor computational performance in most software tools also restrict large-scale applications at the regional or national level. Furthermore, the usage of a strictly microscopic simulation software requires the transport planner to provide detailed inputs about the track layouts, train paths etc., which may not yet be defined in the long-term horizon of strategic studies. Furthermore, most of the existing tools do not allow for incorporating existing or developing new train control mechanisms (train disposition, train management system). These shortcomings collectively underscore the need for more adaptable and open solutions in the domain of railway system simulation.

The objective of this study is to create a straightforward and adaptable open-source tool designed for the microscopic, mesoscopic or hybrid micro-mesoscopic simulation of trains and their interactions with infrastructure. This tool aims to facilitate the identification of conflicts and bottlenecks within the system while also serving as a platform for investigating disturbances and evaluating potential train control strategies. The proposed railway simulation approach provides functionality for a feasibility check and conflict detection of timetables and a given infrastructure. At the same time, the approach allows to investigate large-scale railway concepts at the regional or even national level. Additionally, the modeling approach allows for a detailed consideration of transport demand, which may be relevant in the case of train control and incident management.

The methodological concept is to develop a hybrid micro-mesoscopic simulation approach which extends the agent-based simulation framework MATSim (Multi-Agent Transport Simulation, www.matsim.org) [1]. MATSim is open-source, easy to extend and provides several features to simulate railway operation, in particular the simulation of schedule-based public transit [2, 3]. Most applications of MATSim focus on the simulation of transport demand in which operational aspects of the public transit operator are neglected; transit vehicles move along virtual links and strictly follow a fixed schedule [e.g., 4, 5]. Some studies address operational aspects of the transit schedule in the road transport context, e.g., bus bunching [6]. In the railway context, detailed interaction of transit vehicles has not been considered. In some studies, MATSim is used to address disturbances in the public transit system, however, focusing on the demand reactions and without explicit consideration of operational aspects [7, 8]. In the freight context, MATSim has been used to simulate cargo trains; however, except for the queuing dynamics at terminals, railway-specific operational aspects or train delays are not taken into consideration [9]. MATSim has also been used to address operational aspects of single cargo wagon transport in Switzerland using a network-based routing approach in which, however, the detailed railway-specific dynamics are neglected [10, 11]. The flatland framework is open source and a purely microscopic simulator with a simplified physical representation of the real world which is mainly used to investigate train control strategies by means of Machine Learning [12]. Novotný et al. [13] use a different simulation approach and address the methodological concept of a hybrid micro- and mesoscopic railway simulation approach.

2. Methodology

The developed railway operation approach makes use of the agent-based and dynamic simulation framework MATSim which is briefly described in Sec. 2.1. Sec. 2.2 describes the extension of the MATSim framework. Sec. 2.3 describes the discarded prototype implementation.

2.1. Agent-based simulation framework MATSim

The presented approach uses the simulation framework MATSim [1]. In MATSim, each traveler is simulated individually as a so-called agent. The agents adapt to the transport supply to maximize an individual utility function. In this study, the demand part is less relevant and instead the focus is placed on the transport supply side, including operational aspects.

MATSim's public transit module [2, 3] allows to simulate a given schedule. The transit schedule provides the information about all transit routes. A transit route contains a sequence of stops as well as the connection (sequence of links) on the network between these stops. For each transit route, a route profile describes the departure and arrival time offsets, and a list of departures describes when, how often, and which transit vehicles move along the transit route. A specific transit vehicle is assigned to each departure. Information about the transit vehicle and transit vehicle types are described in a separate vehicle description. One of the key features is that transit vehicles may be delayed, and the actual schedule may deviate from the planned schedule. Transit vehicles may be delayed for various reasons: Boarding and alighting passengers may delay vehicles at stops by extending the dwell time. Queuing effects may delay vehicles between stops. The queuing effects may be caused by any form of vehicle-vehicle interaction, e.g., buses competing for the same infrastructure and blocking each other. In a recent extension, MATSim's public transit module was extended by an implementation in which transit vehicles strictly follow the departure and arrival times given in the transit schedule [14]. That is, workarounds to avoid the delay of transit vehicles (e.g., by increasing the link capacities or running or adding a separate link for each train connection) are no longer necessary.

MATSim's mobility simulation of vehicle-vehicle and vehicle-infrastructure interactions uses a time-step based simulation approach. Each mode is assigned to a specific type of mobility simulation. For road congestion, typically, road segments (links) are simulated as First-In-First-Out queues with a limited outflow rate and storage capacity [15]. An alternative to simulating the detailed queuing effects, agents may also move along virtual links based on a predefined mode speed using the Euclidean distance (teleportation).

2.2. Proposed MATSim extension: Railway simulation module

This section describes how the simulation framework MATSim is extended by the proposed module to simulate railway operation, which is referred to as *railsim*. The *railsim* source code is open source and available as an extension of the MATSim framework: <https://github.com/matsim-org/matsim-lib/tree/master/contribs/railsim> (current version: 2023-10-12, commit 5b3a3f4).

MATSim's pre-existing functionality is modified and adapted to the railway simulation context. The main features are listed below.

- (1) Vehicles have a **spatial dimension** along several infrastructure segments (MATSim links). In the default version of MATSim, a vehicle is only located on exactly one single link (network segment). This logic is now changed, and vehicles may have a spatial extension over more than one link. The front part of a train may be on one link and the back part of the train on another link.
- (2) Vehicles **accelerate and decelerate** based on vehicle- and infrastructure-specific attributes as well as the available capacity ahead of the vehicle.
- (3) The interaction of vehicles as well as the modeling of the infrastructure is adapted to the railway system. Trains are enabled to reserve infrastructure elements (**fixed block segments, moving blocks**), and reservation requests interact with each other. Conflicting infrastructure reservations require some kind of prioritization and train control logic. Moreover, a minimum train headway time accounts for technical restrictions related to the infrastructure.

2.2.1. Implementation

The implementation makes use of a *railsim*-specific **QSim engine** which provides sufficient flexibility to introduce the required train dynamics. Similar to the deterministic public transit approach [14], a predefined mode is handled by its own QSim engine and vehicles of that mode are excluded from the default QSim engine. The *railsim* QSim engine manages the movement of trains and takes into consideration the available capacity of the infrastructure. The *railsim* engine also includes methods for updating train positions, handling departures, blocking and unblocking links, calculating target speeds, and deciding the next update for trains.

To account for technical restrictions of the infrastructure a minimum headway parameter may be defined for each link which is then taken into consideration when moving trains from one link to another.

For the consideration of the infrastructure capacity, MATSim's concept of capacity constrained links is extended towards a resource approach: Capacity effects are modeled at the level of resources. A resource consists of one or several MATSim links and allows for different ways to model the railway infrastructure. As shown in Fig. 1, for each track and for each direction the track can be operated, a MATSim link is generated. Single or multiple MATSim links form resources which are capacity constrained and may be used in different ways to account for fixed blocks, moving blocks or infrastructure elements, such as an intersection. In the microscopic modeling case, the capacity of a resource is 1, that is, a resource may only be occupied by a single train. In the mesoscopic modeling case, the capacity of a resource may be larger than 1 which means that a resource may be used by more than one train simultaneously (see Sec. 2.2.2).

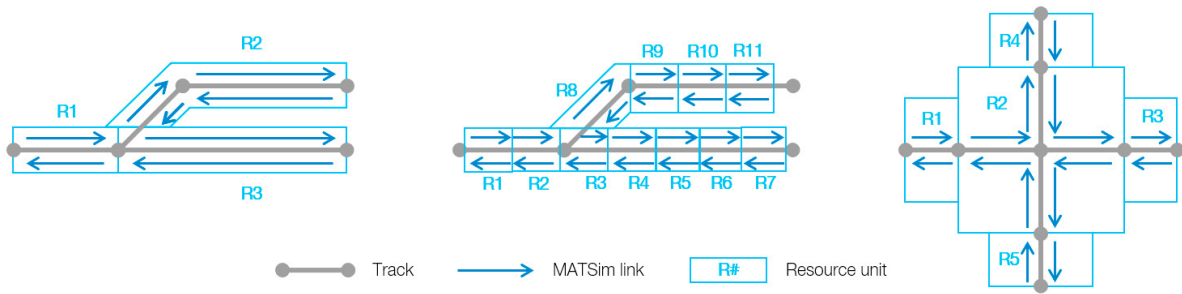


Fig. 1: Implementation of fixed blocks (left), moving blocks (middle), and crossings (right) using the resource concept.

The **speed dynamics** of a train result from the acceleration and deceleration values provided in the vehicle attributes, the maximum velocity of the vehicle, the maximum speed level of the infrastructure and the availability of the infrastructure ahead of the vehicle. The train length is provided in the vehicle attributes and the length of the reserved train path ahead of the train is dynamically computed based on the required braking distance to come to a complete stop or reach a reduced target speed. A train decelerates if a requested downstream resource along the path of the train is blocked by another train or if the target speed on a downstream link is below the current train velocity. That is, the train velocity adjusts to the target speed and available length of the reserved train path. If a train is supposed to stop at a station the reserved train path is cut and does not go beyond that station until the train's next departure, consequently the train decelerates. A train accelerates if the downstream resources are available and the train speed is below the maximum velocity defined by the vehicle type and below the maximum speed level of the requested infrastructure.

In the current version, a simple **train control** functionality is implemented which allows for locally restricted rerouting of trains. Links of the train network may be marked as an entry or exit link for rerouting. If a train then passes along such an entry link a new route is requested from the current link to the first exit link along the original transit route given in the schedule while taking into consideration the available capacity. An example is to use the mesoscopic track segment which leads into a station as an entry link and the mesoscopic track segment behind a station as an exit link and then to model the different tracks in the station at the microscopic level; the train may then adjust its route depending on the available capacity and change from one platform to another one.

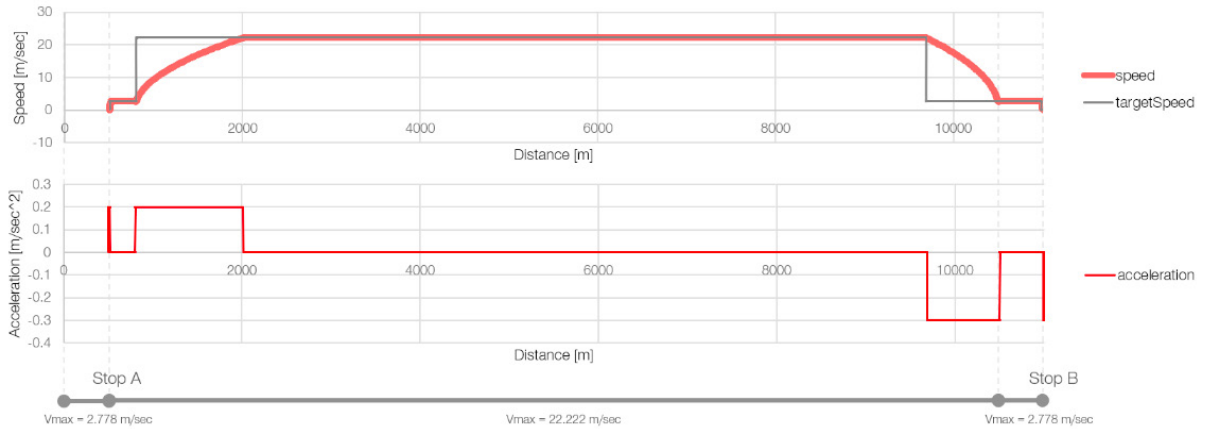


Fig. 2: Simulated dynamics for a train traveling along a simple corridor from stop A to stop B.

Train acceleration = 0.2 m/sec^2 , Train deceleration = 0.3 m/sec^2 . The target speed is an intermediate calculation step for the train dynamics.

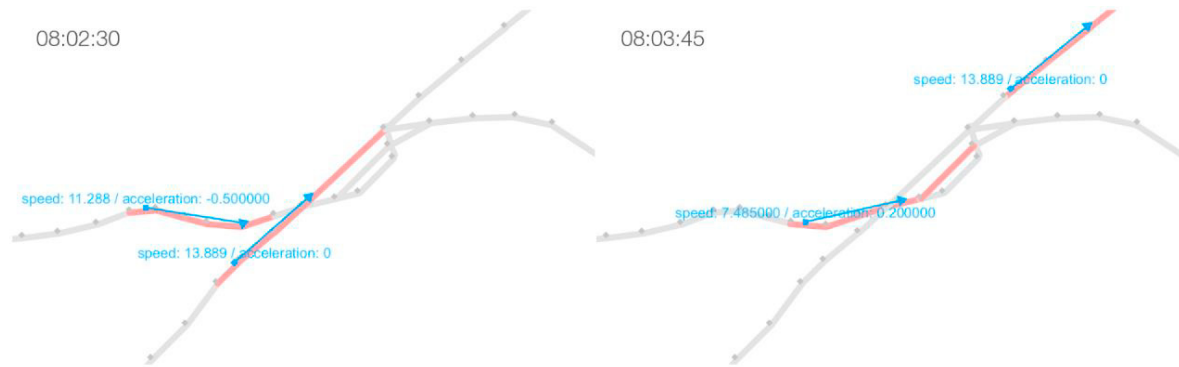


Fig. 3: Microscopic modeling of a train blocked by another train. The infrastructure is drawn in grey (MATSim nodes and links). The spatial expansion of trains is visualized in blue. The reserved infrastructure is marked in red.

In order to remain consistent with the basic principle of MATSim, standard **events**, such as *VehicleEntersLinkEvents* and *VehicleLeavesLinkEvents* are thrown. In addition, other events are thrown that result from the railway-specific context: The *RailsimTrainLeavesLinkEvent* marks that (the end of) the train leaves a link; provides information about the timestamp, the link Id and the vehicle Id. The *RailsimLinkStateChangeEvent* is thrown when state of a link changes, e.g., from “FREE” to “BLOCKED”. The *RailsimTrainStateEvent* is thrown when a train state changes as well as at configurable intervals, e.g., every 10 seconds, and provides information about the head and tail position (link Id and position on the link), the current speed, the target speed and the current acceleration or deceleration of a train. A *RailsimDetourEvent* is thrown when a train was re-routed and provides information about the vehicle and the changed network route.

2.2.2. From the microscopic to the mesoscopic modeling context

The *railsim* contribution is designed in a way that it can be used in a microscopic, mesoscopic or mixed (hybrid) micro- and mesoscopic modeling context. The graphics in Sec. 2.2.1 visualize the microscopic modeling context in which for each track a resource is defined which consists of two MATSim links (one MATSim link per direction). In the mesoscopic modeling context, multiple tracks are considered as one resource with a capacity larger than one.

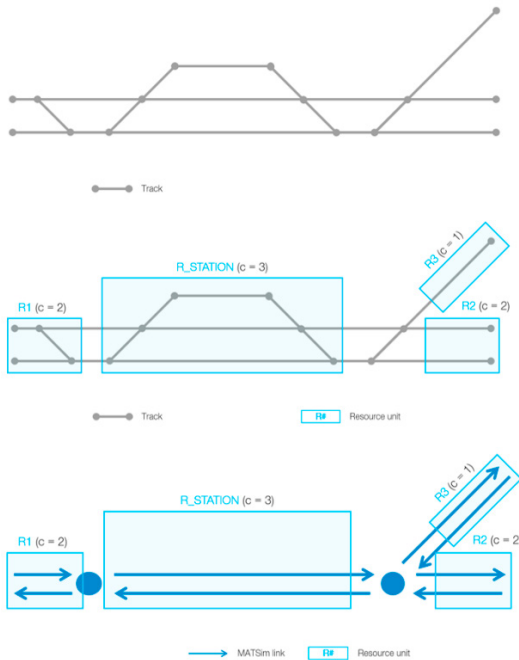


Fig. 4: From a microscopic track layout to a mesoscopic model; c denotes the capacity defined as the maximum number of trains that can simultaneously use the resource.

An example of how to model a track layout at the mesoscopic scale is given in Fig. 4. The mesoscopic modeling approach allows to simplify the complexity of the track layout while at the same time the relevant properties of the infrastructure are taken into consideration. That is, the detailed track layout is parametrized using the resource capacities (maximum number of trains that can simultaneously use the resource). The mesoscopic modeling approach offers several advantages. Instead of the detailed track layouts, only the approximate geometries and capacities or number of tracks are sufficient. This may be useful if exact data is not available, e.g., for studies in which only rough concepts are considered in the long-term planning horizon. A further advantage is that the mesoscopic scale allows for an improved computational performance. The mesoscopic resolution may be used in combination with the microscopic modeling approach. The actual planning area may be modelled microscopically, and the surrounding network may be modelled at the mesoscopic scale. This allows for large-scale application studies in which capacity effects are adequately taken into consideration.

2.3. Prototype implementation using MATSim's signals extension

Before implementing the design described above, MATSim's signals functionality was used in a preliminary prototype. The prototype offers less flexibility and involves longer computation times compared to the final implementation. Nonetheless, this section briefly introduces the basic concept of the prototype implementation. The idea of the prototype was to use MATSim's signals to control the movements of trains. A signal is located at the end of a link and provides a separate signal state for each of the to-links. The to-links are defined as all outgoing links at the to-node of the previous link. The signal state "red" prevents trains to move to the next link and "green" does not restrict a train from moving to the next link. A further idea in the prototype implementation was to consider the transit vehicle as the front of the reserved train path. The position of the actual train is computed backward from the front of the train path. The length of the train path is computed dynamically based on the current speed of the train. An adaptive signal controller controls the movements of trains. Based on the information when train paths enter a link or trains leave a link, in each simulation time step several conditions are checked for each link. If one condition requires red for a from-link and to-link pair, the signal is set to red. If all conditions allow for green for a from-link and to-link pair, the signal is set to green. The following conditions are accounted for: (a) If one or more track segments are blocked by a train itself or a reserved train path and there is no capacity available for any further trains the access to the blocked links is restricted. For all from-links the signal condition is set to red for the blocked to-link. If capacity is available on the to-link, the signal conditions on all from-links are changed back to green. (b) If a link is attributed with a minimum headway time, each time a train leaves a link the signal condition is set to red for all to-links. When the minimum headway time has passed, the signal condition is set back to green for all to-links. (c) In MATSim, links are one-directional. Single direction tracks which can be used in both directions are therefore modeled using two separate links which are connected via their link attributes. If a pair of links is attributed as a one-direction link and a train has entered one of these links the signal condition on the ingoing links of the opposite direction link is set to red. Once the train has left the link the signal for the opposite direction link is set back to green. In addition to the signal conditions described above, further conditions are introduced for rather technical reasons. Conflicts or interactions

within a single time step cannot be captured by MATSim's default signaling logic. To prevent two vehicles from entering the same link in the same time step, the adaptive signal controller accounts for potentially conflicting from-links and alternates the actual green phase between them. A train's maximum velocity is computed as the minimum of the maximum speed value provided in the link attributes and the maximum velocity of the rolling stock. In the prototype implementation, acceleration is only accounted for in a simplistic way (a gradual increase in speed from link to link) and deceleration was not accounted for.

3. Illustrative case study

The developed software extension is applied to a large-scale and mesoscopic modeling context. In a first step, a draft schedule concept for Switzerland in the year 2050 is designed using a network graphic editor. The network graphic editor is an application which is currently closed source and mainly used for long-term planning projects at SBB. The network graphic contains the railway stops and railway lines; for each line, additional information is given about the frequency, product category and the departure and arrival offsets per stop. The network graphic is exported from the network graphic editor application as a JSON file. A network graphic converter was developed to read the JSON file and convert the network graphic into the MATSim format. The converter outputs transit schedule, transit vehicles, and network XML files, which are the necessary input files for running MATSim (see Fig. 5 for a visualization of the network and stops). The converter requires additional data to determine the station coordinates, the route distances between stations and to assign vehicles to route departures.

In a next step, the initial network capacities are systematically modified, and parametric simulation experiments are carried out. The idea of this case study is to explore different infrastructure capacities for the stations as well as for the sections between two stations. To simplify the simulation setup, there is no differentiation in station categories or route section types, instead we use a unique value for all the stations as well as a unique capacity value for track sections. A further assumption is that each section between two stations is defined as a single block segment which can be used by a single train per track. Also, a unique vehicle type is used for the entire schedule. The capacity of the stations is varied in steps from 1 to 15 trains per station. The capacity for the track sections between two stations is varied in steps from 1 to 9 trains per section. In the most extreme case, 9 parallel tracks between any two stations are required. The train schedule itself is fixed in all simulation experiments.

The simulation experiments are analyzed, and the results are compared in terms of their performance. The focus of the comparison is the feasibility of the schedule, which is quantified using the number of trains (successfully) arriving at their destination. The combinatorial results for different assumptions regarding the station or section capacities are shown in Fig. 6. The results reveal that from a capacity of 3 trains per section and 10 trains per station, most trains successfully reach their destination transit stops, and the schedule is considered as feasible. For lower capacities, the infrastructure is not sufficient, trains get stuck and do not reach their destinations. Overall, the results are considered plausible given the assumptions. Further experiments are required in which the station capacity varies among the different stations. Additional simulation experiments may be carried out for different assumptions regarding the length of the block segments, the vehicle type, or more detailed capacities which account for the spatial differences.

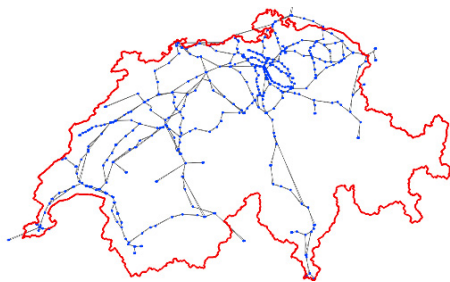


Fig. 5: Rail network (illustrative planning concept)

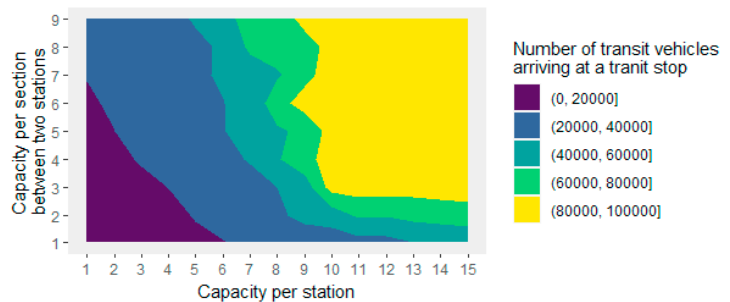


Fig. 6: Simulation results: Feasibility of the train schedule for each combination of station capacity and route section capacity.

4. Conclusion and future directions

In this study, the existing MATSim simulation framework was successfully extended towards a module for the simulation of trains and their interactions with the infrastructure. The proposed railway simulation approach *railsim* offers several advantages over existing railway planning tools: The approach is open-source, easily adaptable, allows for microscopic, mesoscopic or hybrid micro-mesoscopic simulation setups and is fully integrated in an existing simulation framework. The *railsim* module allows to replace the mobility simulation for a predefined mode; the remaining functionality remains fully intact, e.g., multi-modal demand modeling.

The illustrative Switzerland case study reveals that a capacity of 3 trains per section and 10 trains per station ensures most trains reach their destinations successfully, while lower capacities lead to insufficient infrastructure and train disruptions. The implementation was possible without major changes to the MATSim core code. The modeling of the physical layer and the main railway-specific operational elements was straightforward, e.g., fixed block segments, moving blocks, one-directional tracks, train path reservation, train acceleration dynamics, expansion of trains along several infrastructure elements. A somewhat greater challenge was the implementation of train control strategies which is one of the extensions currently in development. In a first step, we focus on a simple deadlock avoidance control strategy. In a next step, more advanced control strategies will be developed and investigated by introducing external disruptions. The existing *railsim* implementation allows for passengers boarding and alighting trains. A further future direction is to add the travel demand layer to a case study and incorporate the passenger volumes into the train control strategies.

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